

Venkatesh Swarna

USA | +1-203-887-0845 | swarnavenkatesh46@gmail.com



Professional Summary

Data Engineer with around 5 years of industry experience designing, building, and supporting production-grade data pipelines and analytics solutions across AWS and Azure environments. Strong expertise in Python, SQL, PySpark with hands-on experience in ETL development, data modelling, workflow orchestration, and automation. Proven ability to collaborate with cross-functional teams, translate business requirements into scalable technical solutions, and improve data reliability, availability, and operational efficiency.

Education

University of New Haven, West Haven, CT

Aug 2023 - May 2025

Masters, Data Science

Professional Experience

Cognizant | Data Engineer - Programmer Analyst

Oct 2021 - Aug 2023

- Designed and operated AWS batch data processing pipelines for structured data sets using Amazon S3, AWS Glue, PySpark, and Spark SQL for data ingestion and transformation for analysis and reporting.
- Created and optimized PySpark & Spark SQL jobs on AWS Glue & EMR, improving performance by utilizing partitioning techniques, predicate push-down, & query optimization.
- Orchestrated data production workflows through triggers in AWS Glue and Amazon CloudWatch, including handling of job failures, data latency, and schema-related errors.
- Worked together with data science and analytics teams to provide cleaned and trustworthy dataset sources in Amazon S3 buckets.
- Ensured data quality checks and validation logic were implemented during the ingestion and transformation stages.
- Supported business reporting by generating data sets that could be queried using Amazon Athena and Redshift.
- Ensure completeness and clarity of documents concerning pipeline logic and schema in a way that facilitates teamwork and development.
- Contributed to project delivery on Agile and Kanban projects, including code reviews, testing, and production support.

Accenture | Application Development Analyst

Dec 2018 - Oct 2021

- Designed and implemented PySpark and Spark SQL pipelines in Azure Databricks, processing 2TB+ of structured and semi-structured data weekly, reducing report generation time and enabling faster enterprise analytics.
- Developed and deployed 4+ interactive Power BI dashboards used by 20+ business users, providing near-real-time insights, reducing manual reporting errors and supporting faster operational decision-making.
- Implemented robust data quality, cleansing, and validation frameworks in PySpark, catching more anomalies before downstream consumption, ensuring reliable and accurate analytics.
- Partnered with pricing, operations, and audit teams to translate business requirements into end-to-end data and BI solutions, accelerating monthly reporting cycles and improving stakeholder confidence in analytics outputs.
- Led unit and regression testing for ETL pipelines in collaboration with QA, ensuring production-ready, high-quality data deliverables by reducing production data errors.

Skills

Programming & Querying: Python, SQL, Java, PySpark

Databases: Amazon RDS, SQL Server, Oracle

Big Data & Data Engineering: Apache Spark, PySpark, ETL/ELT frameworks, Delta Lake

Cloud Platforms: Amazon Web Services (AWS), Microsoft Azure

ETL & Workflow Orchestration: AWS Glue, AWS Step Functions, Azure Data Factory

Analytics & Data Warehousing: Amazon Athena, Amazon Redshift, Azure Synapse Analytics

Data Lake & Storage: Amazon S3, Azure Data Lake Storage Gen2

BI & Visualization: Power BI

Tools & Methodologies: Agile, Kanban, Jira, ServiceNow

Certifications and Achievements

AWS Certified Data Engineer – Associate (DEA-C01)

Microsoft Azure Fundamentals (AZ-900)

Star of the Month Award (multiple times), Accenture — for strong performance and consistent project contributions